(a) Estimate the maximum energy of the β-spectrum. (b) Find the energy of the γ-ray.

7.14. A 40-story high modern business building is supported by 0.9 m thick pillars of reinforced concrete. The insurance company must check that the number of iron bars are as many as required, and therefore they want to investigate the pillars by γ-radiography. What exposure times are required for (a) a small 200 GBq ^{60}Co source, (b) for a large 150 TBq source? Use the same film data as in 7.9.2.

7.15. A swimming-pool reactor produces a flux of 3×10^{16} thermal neutrons $m^{-2}\,s^{-1}$ at 1 m from the reactor center. Assuming a parallel beam of neutrons diffusing up to the surface of the pool where the neutron flux is measured to be 10^8 n $m^{-2}\,s^{-1}$, calculate the thickness (x m) of the water layer required. For thermal neutrons the flux is reduced exponentially with the exponent $x\,L^{-1}$, where L is the diffusion length (2.75 cm in H_2O).

7.16. In a sample of 10.4 TBq of old fission products, the average γ-ray energy is 0.5 MeV and on the average 0.4 γ's are emitted per β-decay. (a) What is the lead shielding required to reduce the γ flux to 10^2 γ $cm^{-2}\,s^{-1}$ at 1.5 m from the source assuming only exponential attenuation? (b) What is the relaxation length? (c) What is the build-up factor?

7.11. LITERATURE

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CHAPTER 8

Radiation Effects on Matter

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Soon after the discoveries of X-rays and radioactivity it was learned that radiation could cause changes in matter. In 1901 P. Curie found that when a radium source was placed on his skin, wounds were produced that were difficult to heal. In 1902 skin cancer was shown to be caused by the radioactivity from radium but 5 years later it was learnt that radium therapy could be used to heal the disease. Large radiation doses were found to kill fungi and microorganisms and produce mutations in plants.

Glass ampoules containing milligrams of radium darkened within a few months and became severely cracked, allowing the leakage of radon gas. In the early years of the investigation of radioactivity, emphasis was on radium and its decay products. Among the radiation effects observed were the fluorescence induced in different salts and the changes in their crystallographic form resulting in colour changes from delocalised charges in the crystals. Metals were found to lose their elasticity and become brittle due to delocalised atoms breaking the inherent structure. Radiation was also found to have a profound effect on the chemical composition of solutions and gases. Water, ammonia and simple organic substances decomposed into more elementary constituents and also combined

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